Verification plan

Project: Memory 16*32

Version: 1.0

Presented by: Abdallah Mohamed salah

Date : 24 sep, 2026

introduction

This document presents the verification plan for the **16×32 Memory** design. It defines the verification objectives, scope, test scenarios, functional coverage, and verification strategy used to ensure that the design meets its functional specifications and operates correctly under normal and corner-case conditions.

IP Design details

The IP is a **16×32 synchronous memory module** consisting of 16 memory locations, each storing 32 bits of data. It provides read and write operations controlled by enable and write-enable signals.

I/O Interface:

- clk: Clock input used to synchronize memory operations.
- rst: Reset input.
- EN: Enables the memory operation.
- WE: Selects the write operation when asserted.
- Address[3:0]: 4-bit address input used to select one of the 16 memory locations.
- Data_in[31:0]: 32-bit input data used during write operations.
- Data_out[31:0]: 32-bit output data read from the selected memory location.
- Valid_out: Indicates that the Data_out value is valid.

Functionality:

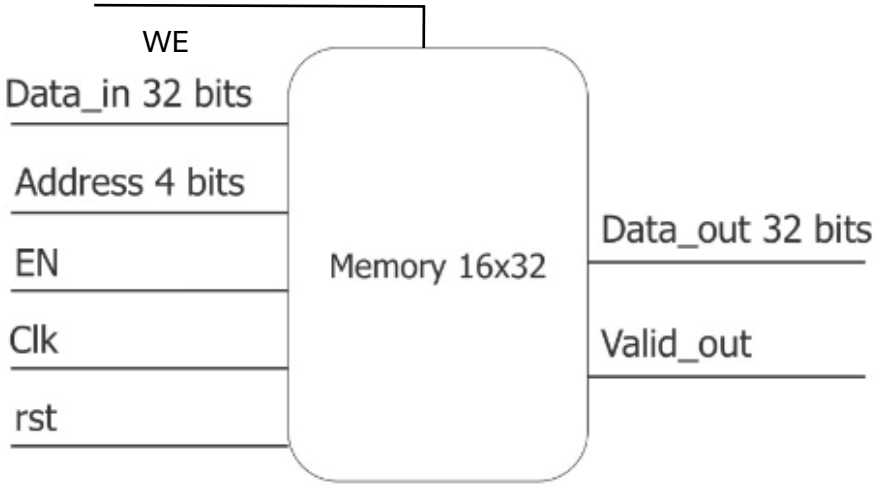
When EN is asserted, the memory performs a read or write operation depending on the state of WE. During a write operation, Data_in is stored at the selected address. During a read operation, the stored data at the selected address is provided on Data_out, with Valid_out indicating valid output data.

Latency:

The memory operations are synchronized to the rising edge of clk. The exact read latency depends on the RTL implementation and the generation of Valid_out.

Capacity:

16 × 32 = **512 bits = 64 Bytes**.



Verification strategy

Used verification strategy is : **module based**

Exit criteria

Verification is considered complete when the following criteria are met:

- **100% of planned functional test cases** are executed.
- All **critical and high-priority test cases pass**.
- **Functional coverage reaches 100%** for all defined verification scenarios.
- All identified **critical/high-severity bugs are resolved and verified**.
- No unexplained or unexpected test failures remain.
- All required **corner cases** have been verified, including boundary addresses and reset behavior.

- The RTL meets the defined functional requirements with no outstanding verification blockers.

Test items

Item shall be :e

- Rst : Active low reset.
  - Low : Asserted.
  - High : Deasserted.
- EN: Enables the memory operation.
  - High : Enable read or write.
  - Low : Disable both read and write.
- WE: select between read and write operation.
  - High : write.
  - Low : read.
- Address[3:0]: 4-bit address input used to select one of the 16 memory locations.
  - Value between 0 : 15
- Data_in[31:0]: input to be written into memory
  - Can be any value from 0 : 4G (4 294 967 295)

Test case

ID	Test Case Name	Description
TC01	Reset_Test	Verify that the memory enters the expected reset state when rst is asserted.
TC02	Basic_Write_Test	Verify that 32-bit data can be successfully written to a selected memory address.
TC03	Basic_Read_Test	Verify that previously stored data can be correctly read from a selected memory address.
TC04	Write_Read_Test	Verify that data written to a memory location can be correctly read back from the same location.
TC05	All_Address_Test	Verify read and write operations across all 16 memory addresses from 0 to 15.

TC06	Boundary_Address_Test	Verify correct operation at the minimum and maximum addresses, 0 and 15.
TC07	Address_Isolation_Test	Verify that writing to one memory location does not affect data stored in other locations.
TC08	Overwrite_Test	Verify that writing new data to an already occupied address correctly replaces the previously stored data.
TC09	Zero_Data_Test	Verify correct storage and retrieval of 32'h00000000.
TC10	All_Ones_Data_Test	Verify correct storage and retrieval of 32'hFFFFFFFF.
TC11	Data_Pattern_Test	Verify memory operation using different data patterns such as AAAA, 5555, and other representative patterns.
TC12	Back_to_Back_Write_Test	Verify consecutive write operations to different memory locations.
TC13	Back_to_Back_Read_Test	Verify consecutive read operations from different memory locations.
TC14	Read_Write_Transition_Test	Verify correct behavior when transitioning between read and write operations.
TC15	Repeated_Address_Test	Verify correct behavior when performing multiple consecutive operations on the same address.
TC16	Ascending_Address_Test	Verify memory operations using addresses in ascending order from 0 to 15.
TC17	Descending_Address_Test	Verify memory operations using addresses in descending order from 15 to 0.
TC18	Random_Address_Data_Test	Verify memory functionality using randomized addresses and 32-bit data values.
TC19	Enable_Disable_Test	Verify that no memory operation occurs when EN is deasserted.
TC20	Enable_Write_Combinations_Test	Verify all valid combinations of EN and WE and their corresponding behavior.
TC21	Reset_During_Write_Test	Verify the memory behavior when reset is asserted during a write operation.

TC22	Reset_During_Read_Test	Verify the memory behavior when reset is asserted during a read operation.
TC23	Reset_Recovery_Test	Verify that the memory correctly resumes normal operation after reset is deasserted.
TC24	Read_Validity_Test	Verify that Valid_out is asserted correctly when Data_out contains valid read data.